

# RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments

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THESIS RESEARCH PLAN



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# Resumo

PUT HERE ABSTRACT IN PORTUGUESE.

**Palavras-chave:** localização e mapeamento simultâneo, robô móvel autônomo, autonomia a longo prazo, ambientes dinâmicos.



# Abstract

PUT HERE ABSTRACT IN PORTUGUESE.

**Keywords:** Simultaneous Localization and Mapping (SLAM), Autonomous Mobile Robot (AMR), long-term autonomy, lifelong SLAM, dynamic environments.



# Contents

|                                                                                                            |            |
|------------------------------------------------------------------------------------------------------------|------------|
| <b>Resumo</b>                                                                                              | <b>i</b>   |
| <b>Abstract</b>                                                                                            | <b>iii</b> |
| <b>Abbreviations</b>                                                                                       | <b>xi</b>  |
| <b>1 Introduction</b>                                                                                      | <b>1</b>   |
| 1.1 Context . . . . .                                                                                      | 1          |
| 1.2 Motivation . . . . .                                                                                   | 3          |
| 1.3 Goals . . . . .                                                                                        | 4          |
| 1.4 Research Methodology . . . . .                                                                         | 4          |
| 1.5 Document Organization . . . . .                                                                        | 5          |
| <b>2 Literature Review</b>                                                                                 | <b>7</b>   |
| 2.1 Research Methodology . . . . .                                                                         | 7          |
| 2.2 Simultaneous Localization and Mapping (SLAM) . . . . .                                                 | 7          |
| 2.3 Long-term SLAM . . . . .                                                                               | 7          |
| 2.4 Summary . . . . .                                                                                      | 7          |
| <b>3 Thesis Statement</b>                                                                                  | <b>9</b>   |
| 3.1 Problem Statement . . . . .                                                                            | 9          |
| 3.2 Research Questions and Hypotheses . . . . .                                                            | 9          |
| 3.3 Aim and Scope . . . . .                                                                                | 9          |
| 3.4 Summary . . . . .                                                                                      | 9          |
| <b>4 Preliminary Developments</b>                                                                          | <b>11</b>  |
| 4.1 Robot Calibration . . . . .                                                                            | 11         |
| 4.2 Introduction to SLAM . . . . .                                                                         | 11         |
| 4.3 Summary . . . . .                                                                                      | 11         |
| <b>5 Research Plan</b>                                                                                     | <b>13</b>  |
| 5.1 Tasks and Milestones . . . . .                                                                         | 13         |
| 5.2 Timeline . . . . .                                                                                     | 13         |
| 5.3 Scientific Dissemination . . . . .                                                                     | 13         |
| 5.4 Summary . . . . .                                                                                      | 13         |
| <b>References</b>                                                                                          | <b>15</b>  |
| <b>A A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots (preprint)</b> | <b>19</b>  |



# List of Figures

|     |                                                                                                                          |   |
|-----|--------------------------------------------------------------------------------------------------------------------------|---|
| 1.1 | Global Autonomous Mobile Robot market estimated growth by AgileIntel Research during 2018–2029 in US\$ million . . . . . | 2 |
| 1.2 | Process model of the Design Science Research Methodology (DSRM) . . . . .                                                | 5 |



# List of Tables



# Abbreviations

**AI** Artificial Intelligence. 3

**AMR** Autonomous Mobile Robot. 1–4

**ARMA** Auto-regressive Moving Average Model. 3

**BDVA** Big Data Value Association. 3

**CAGR** Compound Annual Growth Rate. 1

**CLAIRE** Confederation of Laboratories for Artificial Intelligence Research in Europe. 3

**CNN** Convolutional Neural Network. 2

**DSRM** Design Science Research Methodology. 4, 5

**ELLIS** European Lab for Learning & Intelligent Systems. 3

**EurAI** European Association for Artificial Intelligence. 3

**euRobotics** European Robotics. 3

**FreME**n Frequency Map Enhancement. 3

**LiDAR** Light Detection And Ranging. 2

**LTM** Long-Term Memory. 3

**POV** Point-Of-View. 4

**SDG** Sustainable Development Goal. 4

**SLAM** Simultaneous Localization and Mapping. 1–5

**SRIDA** Strategic Research, Innovation and Deployment Agenda. 3, 4

**STM** Short-Term Memory. 3



# Chapter 1

## Introduction

This research aims to contribute to the progress of the current state of the art on **Simultaneous Localization and Mapping (SLAM)**. This work focuses on challenges inherent to the long-term operation of **Autonomous Mobile Robots (AMRs)** while performing SLAM in dynamic indoor environments. The main goals are to maintain a stable and up-to-date representation of the environment and, consequently, improve the localization performance of **AMRs**. In this context, the proposed approach should develop new algorithms and take advantage of emerging technologies to overcome the challenges found through the research work.

This chapter is divided into five sections. Section 1.1 contextualizes this research work and describes the primary challenges related to long-term **Simultaneous Localization and Mapping (SLAM)**. Section 1.2 presents the motivations that led to the formulation of this research plan. Then, Section 1.3 establishes the goals for the PhD thesis. Section 1.4 details the research methodology that will support the research work. Finally, Section 1.5 defines the structure of this document.

### 1.1 Context

Nowadays, logistic and retail companies, warehouses, healthcare institutions, automotive and aerospace industries, agriculture businesses, and healthcare institutions are using **AMRs** to automate tasks such as transportation and security monitoring. The tasks automation improves operational efficiency, flexibility, and productivity achieving higher efficiencies and lower production costs [1]. Consequently, these improvements contribute to the growth of the global economy and increase the global interest in the **AMR** market. Indeed, a study elaborated by AgileIntel Research [2] estimates the value of the global **AMR** market at US\$2,525.66 million in 2021 and is estimated to grow at a **Compound Annual Growth Rate (CAGR)** of 15.4% between 2022 and 2029, as illustrated in Figure 1.1. The conclusions of the study highlight the importance of **AMRs** for global economics while also forecast an increasing demand for **AMR** systems in the following years from 2022 [2].

An **AMR** requires a representation of its surroundings by extracting meaningful data from its sensors to localize itself relative to the environment [3]. **SLAM** addresses this problem by incorporating the robot state estimation concurrently with the mapping process to build a representation of the environment perceived by the robot. The state is described by the robot's pose (position and orientation), even though other variables may be included in the state, such as velocity and calibration parameters. As for the environment representation, the mapping process originates a map incrementally built when exploring unknown areas or refined on passages through known locations in order to describe the environment in which the robot operates [4].

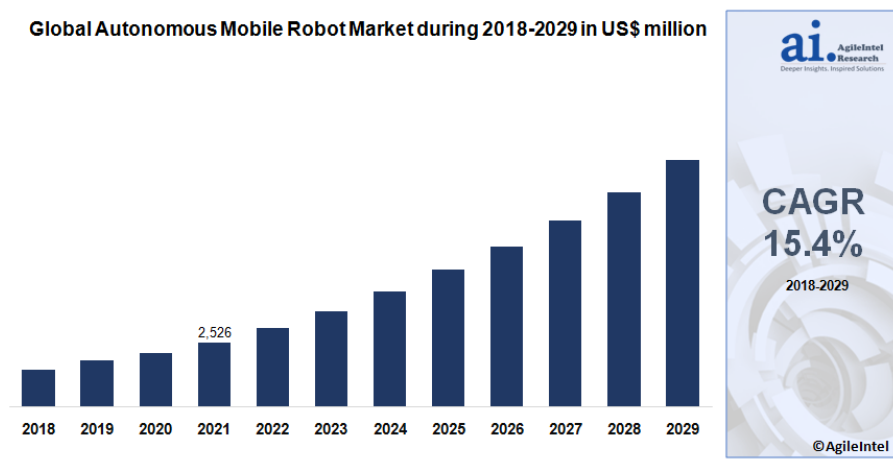


Figure 1.1: Global **Autonomous Mobile Robot (AMR)** market estimated growth by AgileIntel Research during 2018–2029 in US\$ million [2]

In a static scene, the robot would only need to map once because the map would remain consistent with the environment throughout time. However, industrial locations, agricultural sites, warehouses, and service-oriented applications such as shopping centers and healthcare institutions have inherently moving elements in the scene (humans, objects), reconfiguration of the environment (reorganization of machines, patient beds), and appearance variations (lighting, weather, seasonal, or daytime changes). The deployment of **AMRs** in these varying conditions are a challenge for the **SLAM** system, where the system should decide how (consider the most current state, the evolution of the environment dynamics, only the most static changes?) and when to update the map (when the variations occur, after a certain time?). This challenge is also known as the stability-plasticity dilemma, where a lifelong **SLAM** system should both adapt to new environment changes and preserve old states over time [5]. Furthermore, the map would grow indefinitely when gathering new information from the environment without any map management system. This ever-growing problem poses another challenge for the **SLAM** system in the long-term due to the limited computational resources of the mapping vehicle. Indeed, the map size should be dependent on the environment area and not on the trajectory length of the mapping process nor operation time, only growing when the robot would explore unknown locations [6].

Currently, the literature is predominantly focused on the appearance invariance challenge of long-term localization and mapping, especially in outdoor environments, given the recent growing interest in autonomous vehicles. Although vision sensors are greatly affected by appearance variations, **Convolutional Neural Network (CNN)**-based features extracted from the visual images are being used for localization in challenging outdoor scenes, given their more discriminative properties compared to handcrafted features [7–12]. The current research also uses other sensor modalities, such as **Light Detection And Ranging (LiDAR)** [13–16] (also known as 3D laser scanners) and Radar [16, 17], to overcome some limitations of vision sensors. While visual data is affected by illumination changes of day-night situations or changing weather and seasonal conditions, laser-based localization does not suffer from illumination variance. Even so, fog or direct light can still affect the laser readings. As for radar, the sensor is invariant to both lighting and weather changes. Still, noisy measurements affect the performance of radar-based **SLAM** in long-term scenarios [16]. Moreover, the existent literature also evaluates the long-term stability of features to only using stable ones for localization and mapping, using as scoring functions predictors such as number of observations and concentration ratio [18, 19].

Several works already focus on the ever-growing problem of continuously performing **SLAM**. The concept of mutual information from information theory is the most common methodology to determine which nodes to remove from the map for the graph formulation of **SLAM** [6, 20–23]. However, this methodology is mostly intended for offline execution (*e.g.*, between running sessions) due to its additional computational cost when operating online. Other methodologies to manage the size of the graph evaluate outdated information between current view and the map [24, 25], the spatial density of nodes [26, 27], and the node time recency [28, 29]. In feature-based **SLAM**, management policies such as counting matching observations of features [30, 31] and using multiple predictor functions [18, 19, 32, 33], similarly to appearance invariance methods, also try to constraint the number of mapped features to only be dependent on the explored environment area.

As for dealing with dynamic environments, an intuitive approach is to distinguish between static and dynamic observations, favoring the stable ones for localization, and having two maps representing the most stable and the more dynamic representations of the environment [24, 34, 35]. Inspired by the human memory multi-store model of Atkinson and Shiffrin [36], **Short-Term Memory (STM)** and **Long-Term Memory (LTM)** maps try to capture at least two different levels of dynamics [5, 30, 31]. Furthermore, the rejection of dynamic objects or features for localization and mapping to only consider stable observations is performed in the literature by methods such as the YOLO network [35, 37] and pixel-wise semantic segmentation [38–40]. The methodologies mentioned previously for dealing with dynamic environments model the environment dynamics as static (permanent change), semi-static (short-term static object or static at the current observation instant), or dynamic (moving object in the scene). In contrast, assuming the independence of observations, Markov-based processes can be used to model and predict a value for the environment dynamics [41, 42]. However, these models are usually intended for offline usage between sessions and the assumption of independent observations may not be valid because static and dynamic objects may influence the dynamics of their surroundings. Although the **Frequency Map Enhancement (FreME)** [43] algorithm estimates online the dynamics through spectral analysis, the method only assumes periodic changes in the environment. The **Auto-regressive Moving Average Model (ARMA)** [44] models both aperiodic and periodic changes but, still, its offline operation does not allow an online estimation of the environment dynamics.

## 1.2 Motivation

A joint partnership between major European activities for **Artificial Intelligence (AI)** – **Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE)**, **European Lab for Learning & Intelligent Systems (ELLIS)**, and **European Association for Artificial Intelligence (EurAI)** –, **Data – Big Data Value Association (BDVA)** –, and **Robotics – European Robotics (euRobotics)** – released a **Strategic Research, Innovation and Deployment Agenda (SRIDA)** [45] in September 2020. This agenda identifies key and critical technical areas for the future of Robotics, among them are dealing with unpredictable and unknown environments and irreversible actions. Indeed, in many service applications, surrounding environments of autonomous systems usually involve the interaction with humans. Furthermore, bad policies can bring autonomous systems to unrecoverable states from which learning is no longer possible, requiring the environment to be reset to start the process again. For example, the influence of dynamic elements in the process of **SLAM** with **AMRs** can make the environment representation saved in the robot outdated and unusable due to changes in the scene without any module to update the map accordingly to the environment changes. An outdated map representation would

required a specialized operator to remap the environment, possibly redefining the trajectories if needed, and then restart the autonomous system operation.

Subsequently, the interpretation and processing of dynamic changes in the environment by **AMRs** would enable the maintenance of an up-to-date map representation of the robots' surroundings. Robot systems would be able to better understand their operating environment, which may include humans, while also improving the efficiency and safety of these systems. For these outcomes, **SRIDA** [45] also identifies technology and perception enablers such as robot systems being able to interpret sensor data in real-time – particularly in complex environments or fusion of multi-modal sensor data (vision, ranging, acoustic) to increase self-awareness of those systems –, matching local **Point-Of-View (POV)** with world views, and full 3D perception systems and sensors able to decompose and interpret whole scenes in real-time to 4D.

### 1.3 Goals

The main scope of this research work, *RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments*, is performing lifelong **SLAM** within dynamic environments with an **AMR**. The focus is developing map maintenance algorithms capable of handling dynamic changes and moving elements in the operating area of autonomous robot systems. Thus, the main goals of this research proposal aiming to contribute to the progress of the current state of the art on **SLAM** are the following ones:

- modeling of environment dynamics perceived by multi-modal sensors (2D and 3D ranging data, vision) in the map representation;
- account the estimated values of the environment dynamics incorporated in the map for both localization and mapping processes;
- achieve long-term localization and mapping in dynamic environments.

Within the scope of the 2030 Agenda for sustainable development agreed by 195 nations in September 2015 [46], this research work falls under the following **Sustainable Development Goal (SDG)**:

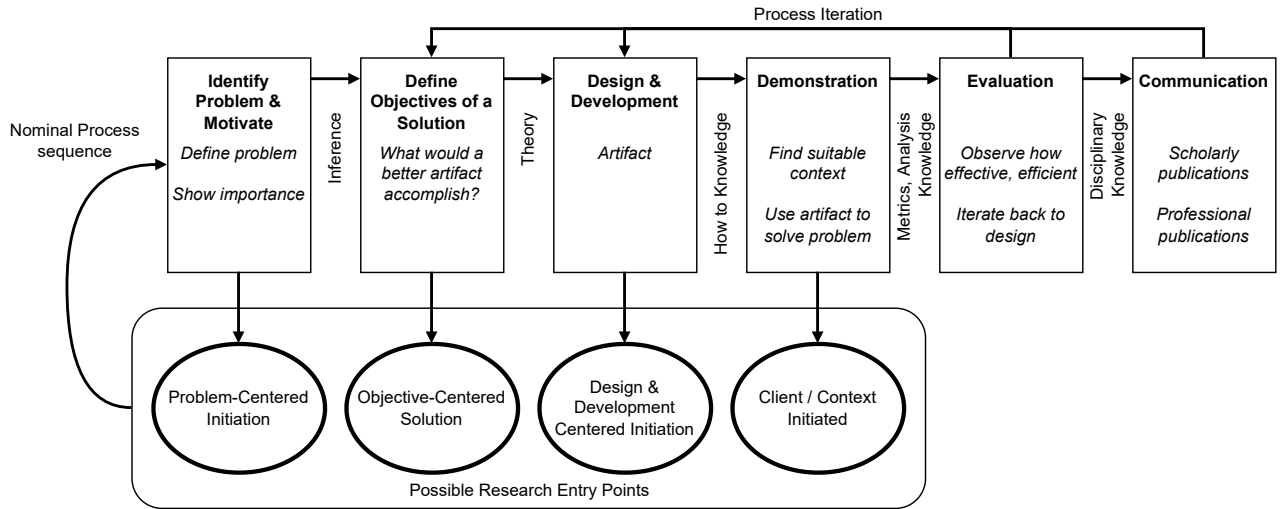
- SDG8: Decent work and economic growth;
- SDG9: Industry, innovation and infrastructure;
- SDG12: Responsible Consumption and Production.

### 1.4 Research Methodology

The definition of a base research methodology is crucial to formulate a robust research strategy for obtaining successful results. In engineering disciplines, the **Design Science Research Methodology (DSRM)** is considered a valid and valuable method to obtain applicable and effective problem solutions incrementally [47]. Thus, this PhD thesis will follow a **DSRM** strategy as the one presented in Peffers *et. al.* [47] and illustrated in Figure 1.2. **DSRM** incorporates the principles, practices, and procedures to carry out research projects. The methodology consists on the following nominal sequential processes:

1. Problem identification and motivation;
2. Definition of the objectives for a solution;
3. Design and development;
4. Demonstration;
5. Evaluation;

## 6. Communication.

Figure 1.2: Process model of the **Design Science Research Methodology (DSRM)** [47]

The first process defines the specific research problem and justifies the value of a solution and its relevance to the scientific community. The second step defines both quantitative (based on key performance indicators and in comparison to existing solutions) and qualitative (functional description of the solution) goals from the problem definition. The following step focuses on design and development activities for creating an artifact, defining its desired functionality and architecture and then actually creating it. In the scope of this PhD thesis, an artifact can be models, algorithms, methodologies, among others. The *Demonstration* activity uses the artifact to solve one or more scenarios or case studies of the problem. Then, the next process is the observation and measurement of how well the artifact supports the solution of the problem. The *Evaluation* can be a comparison of the artifact's functionality with the objectives from the second activity in the **DSRM** model, simulations, or feedback from the end-user of the solution. Lastly, the *Communication* step focuses on the scientific dissemination of the problem and its importance, the artifact, its novelty and utility, the methodology for its design, and its effectiveness to researchers and other relevant audiences such as specialized operators. Although Peffers *et. al.* [47] structures the **DSRM** model in a nominally sequential order, the methodology does not obligate the researchers to always proceed sequentially from the first to the sixth activity. Indeed, Peffers *et. al.* [47] defines possible research entry points for the researchers to begin and then move outward.

## 1.5 Document Organization

The remaining of this document is organized as follows. Chapter 2 presents a literature review on concepts and technologies related to the topics of **SLAM** and long-term localization and mapping in dynamic environments. Chapter 3 provides the problem statement and the main goals and scope of this research proposal. The statement describes the problem and formulates the research questions that should be answered during the execution of this research work and possible hypotheses that support potential answers to those questions. Chapter 4 describes preliminary work and developments. Finally, Chapter 5 outlines the work plan defining the tasks and milestones predicted for the research and the respective timeline. This chapter also discusses the scientific dissemination methodology that will be followed during the execution of the research plan.



## **Chapter 2**

# **Literature Review**

### **2.1 Research Methodology**

### **2.2 Simultaneous Localization and Mapping (SLAM)**

### **2.3 Long-term SLAM**

### **2.4 Summary**



## **Chapter 3**

# **Thesis Statement**

### **3.1 Problem Statement**

### **3.2 Research Questions and Hypotheses**

### **3.3 Aim and Scope**

### **3.4 Summary**



## **Chapter 4**

# **Preliminary Developments**

### **4.1 Robot Calibration**

### **4.2 Introduction to SLAM**

### **4.3 Summary**



## **Chapter 5**

# **Research Plan**

### **5.1 Tasks and Milestones**

### **5.2 Timeline**

### **5.3 Scientific Dissemination**

### **5.4 Summary**



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## **Appendix A**

# **A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots (preprint)**

Ricardo B. Sousa, Héber M. Sobreira, and António Paulo Moreira. “A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots” [Manuscript submitted for publication]. 2022. Department of Electrical and Computer Engineering, Faculty of Engineering of the University of Porto (FEUP). Centre for Robotics in Industry and Intelligent Systems (CRIIS), INESC TEC – Institute for Systems and Computer Engineering, Technology and Science. URL: <https://github.com/sousarbarb/slr-ltln-mr/blob/main/preprint/pdf.pdf>